THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest 4-digit number divisible by 13 is: 1

(a) 1002 **(b)** 1001 **(c)** 988 **(d)** 1014

2. The number of composite numbers from 1 to 30 (both included) is: 1

(a) 18 **(b)** 20 **(c)** 21 **(d)** 19

3. The smallest number divisible by each of 6, 8 and 12 is: 1

(a) 23 **(b)** 24 **(c)** 25 **(d)** 30

4. The greatest number that divides 48, 72 and 120 leaving no remainder is: 1

(a) 25 **(b)** 23 **(c)** 24 **(d)** 26

5. For the 3-digit number 1d3, the digit d that makes it divisible by 9 (smallest such d) is: 1

(a) 14 **(b)** 6 **(c)** 7 **(d)** 5

6. Which is the greatest 3-digit number that leaves remainder 3 when divided by 7? 1

(a) 997 **(b)** 1004 **(c)** 998 **(d)** 990

7. The sum of the first 3 prime numbers after 10 is: 1

(a) 43 **(b)** 42 **(c)** 40 **(d)** 41

8. The largest prime number less than 100 is: 1

(a) 96 **(b)** 97 **(c)** 99 **(d)** 98

9. **Assertion (A):** The number 1 is neither prime nor composite. 1
Reason (R): The number 1 has only one factor, namely itself.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** 2 is the only even prime number. 1
Reason (R): Every even number greater than 2 has 2 as a factor other than 1 and itself.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the smallest number divisible by 8, 12 and 18. 2

12. Show that the product of 3 consecutive numbers starting from 14 is divisible by 3. 2

13. Find the prime factors of 84 and the smallest number to multiply to make it divisible by 45. 2

14. Find the smallest number divisible by 10, 14 and 35. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Three traffic lights change every 48, 72 and 108 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

16. Find the greatest number that divides 76 and 100 leaving remainder 4 in each case. 3

17. Find the smallest number which when divided by 6, 8 and 12 leaves remainder 5 each time. 3

OR

Three traffic lights change every 30, 45 and 60 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

18. Find the greatest number that divides 232 and 280 leaving remainder 4 in each case. 3

OR

Find the smallest number which when divided by 10, 12 and 15 leaves remainder 3 each time. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Two temple bells ring together at 9:00 am. One bell rings every 12 minutes and the other every 18 minutes. 5

(i) Write 12 and 18 as products of prime factors.

(ii) Find the LCM of 12 and 18.

(iii) At what time will they ring together next?

(iv) How many more times will they ring together before 11:00 am? Give the times.

20. A rectangular garden is 72 m by 48 m. Square flower beds of the largest possible equal size are to be made covering the garden. Find 5

(i) the side of each bed,

(ii) the number of beds,

(iii) the cost of fencing all beds at ₹4 per metre if each bed is fenced separately.

OR

Find the smallest 4-digit number divisible by 10, 15 and 18. Also find the largest 4-digit number divisible by all three. 5

END OF QUESTION PAPER — Total: 40 Marks